



آمار و احتمال مهندسی

Engineering Probability & Statistics

Lecture Notes ▪ جزوه درسی

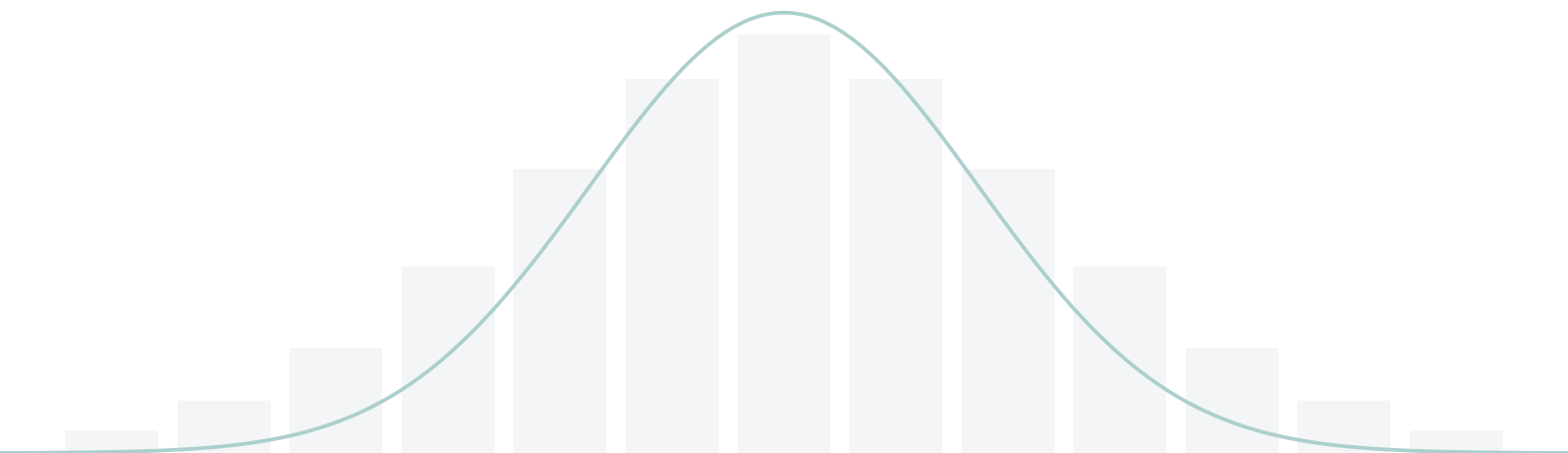
Module 15

Hypothesis Testing

آزمون فرض

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Learning Objectives

After completing this module, students will be able to:

1. Formulate null and alternative hypotheses for engineering problems
2. Identify Type I and Type II errors and their engineering consequences
3. Compute test statistics and p-values
4. Perform Z-tests and t-tests for means
5. Correctly interpret p-values (and avoid common misconceptions)
6. Choose between one-sided and two-sided tests

15.1 Introduction: Making Decisions from Data

Engineering question: “Is this sensor biased?” “Did the process change?” “Does the new algorithm perform better?”

We need a **systematic framework** to answer such questions from data, while controlling the probability of making errors.

15.2 Framework

■ Null Hypothesis H_0

The **default assumption** — typically “nothing changed” or “no effect”:

- ▶ $H_0: \mu = \mu_0$ (sensor is not biased)
- ▶ $H_0: \mu_1 = \mu_2$ (two methods perform equally)
- ▶ $H_0: \sigma^2 = \sigma_0^2$ (variance hasn't changed)

■ Alternative Hypothesis H_1

What we want to **detect** or prove:

- ▶ $H_1: \mu \neq \mu_0$ (sensor IS biased — two-sided)
- ▶ $H_1: \mu > \mu_0$ (performance improved — one-sided)
- ▶ $H_1: \sigma^2 > \sigma_0^2$ (variance increased)

■ Decision

Based on data, either:

- ▶ **Reject H_0** → evidence supports H_1
- ▶ **Fail to reject H_0** → insufficient evidence against H_0

▲ “Fail to reject H_0 ” $\neq$ “ H_0 is true”. Absence of evidence $\neq$ evidence of absence.

15.3 Types of Errors

	H_0 True (reality)	H_1 True (reality)
Reject H_0 (decision)	Type I Error (α)	Correct! (Power = $1-\beta$)
Fail to reject H_0 (decision)	Correct!	Type II Error (β)

■ Type I Error (False Alarm) — probability α

Rejecting H_0 when it's actually true. "Crying wolf."

Engineering analogy: Declaring a process has changed when it hasn't → unnecessary recalibration.

■ Type II Error (Missed Detection) — probability β

Failing to reject H_0 when H_1 is actually true. "Missing the signal."

Engineering analogy: Failing to detect a sensor bias → biased measurements continue.

■ Engineering Analogy: Radar Detection

Statistical Term	Radar Equivalent
H_0 : no target	No aircraft present
H_1 : target present	Aircraft present
Type I error (α)	False alarm
Type II error (β)	Missed detection
Power ($1-\beta$)	Probability of detection

■ The Tradeoff

Reducing α (fewer false alarms) → increases β (more missed detections), and vice versa.

Fix α (significance level), then maximize power = $1 - \beta$.

15.4 Significance Level α

The **maximum acceptable Type I error rate**, chosen BEFORE looking at data.

Common choices: $\alpha = 0.05, 0.01, 0.10$

Choosing α : Consider the cost of a false alarm:

- ▶ $\alpha = 0.01$: Very conservative (e.g., medical device safety)
- ▶ $\alpha = 0.05$: Standard in most engineering/science
- ▶ $\alpha = 0.10$: More tolerant (exploratory analysis)

15.5 Power of a Test

$$\text{Power} = 1 - \beta = P(\text{reject } H_0 | H_1 \text{ is true})$$

Power depends on:

1. **Effect size:** Larger true difference $\rightarrow$ easier to detect $\rightarrow$ more power
2. **Sample size n :** More data $\rightarrow$ more power
3. **Significance level α :** Larger $\alpha \rightarrow$ more power (but more false alarms)
4. **Variability σ :** Less noise $\rightarrow$ more power

15.6 The p-value

Definition

The **p-value** is the probability of obtaining a test statistic **as extreme or more extreme** than the observed value, **assuming H_0 is true**.

$$p\text{-value} = P(\text{data this extreme or more} | H_0 \text{ true})$$

Decision Rule

- ▶ If $p\text{-value} < \alpha \rightarrow \text{Reject } H_0$
- ▶ If $p\text{-value} \geq \alpha \rightarrow \text{Fail to reject } H_0$

✓ CORRECT Interpretation

“The p-value is the probability of seeing data this extreme (or more) if H_0 were true. A small p-value means the data is unlikely under H_0 .”

✗ WRONG Interpretation

“The p-value is the probability that H_0 is true.”

Why wrong: H_0 is either true or false (fixed). The p-value says nothing about $P(H_0 \text{ true})$. It measures how surprising the data is under H_0 .

Calibration

p-value	Evidence against H_0
> 0.10	Weak or none
$0.05 - 0.10$	Marginal
$0.01 - 0.05$	Moderate
$0.001 - 0.01$	Strong
< 0.001	Very strong

15.7 One-Sided vs. Two-Sided Tests

■ Two-Sided ($H_1: \mu \neq \mu_0$)

Used when deviation in **either** direction is important.

- ▶ Reject if $|T| > t_{\{\alpha/2, n-1\}}$
- ▶ p-value = $2 \cdot P(T > |t_{\text{obs}}|)$

■ One-Sided ($H_1: \mu > \mu_0$ or $H_1: \mu < \mu_0$)

Used when only one direction of deviation matters.

- ▶ $H_1: \mu > \mu_0$: Reject if $T > t_{\{\alpha, n-1\}}$
- ▶ $H_1: \mu < \mu_0$: Reject if $T < -t_{\{\alpha, n-1\}}$

■ When to Use Which

Scenario	Test Type
Is the sensor biased (either direction)?	Two-sided
Does the new algorithm IMPROVE BER?	One-sided ($H_1: \text{BER} < \text{BER}_{\text{old}}$)
Has noise power INCREASED?	One-sided ($H_1: \sigma^2 > \sigma_0^2$)
Did the process mean CHANGE?	Two-sided

15.8 Tests for Means

■ One-Sample Z-Test (σ known)

Test: $H_0: \mu = \mu_0$

Test statistic: $Z = (\bar{X} - \mu_0)/(\sigma/\sqrt{n})$

Under H_0 : $Z \sim N(0,1)$

■ One-Sample t-Test (σ unknown)

Test: $H_0: \mu = \mu_0$

Test statistic: $T = (\bar{X} - \mu_0)/(S/\sqrt{n})$

Under H_0 : $T \sim t(n-1)$

■ Two-Sample t-Test (comparing two means)

Test: $H_0: \mu_1 = \mu_2$

Test statistic: $T = (\bar{X}_1 - \bar{X}_2) / \sqrt{(S_1^2/n_1 + S_2^2/n_2)}$

Approximate df by Welch-Satterthwaite formula.

15.9 Test for Variance (Chi-Square Test)

■ One-Sample Test

$$H_0: \sigma^2 = \sigma_0^2$$

$$\text{Test statistic: } \chi^2 = (n-1)S^2/\sigma_0^2$$

$$\text{Under } H_0: \chi^2 \sim \chi^2(n-1)$$

■ Decision

- ▶ $H_1: \sigma^2 > \sigma_0^2$: Reject if $\chi^2 > \chi^2_{\alpha, n-1}$
- ▶ $H_1: \sigma^2 \neq \sigma_0^2$: Reject if $\chi^2 < \chi^2_{1-\alpha/2, n-1}$ or $\chi^2 > \chi^2_{\alpha/2, n-1}$

15.10 Step-by-Step Hypothesis Testing Procedure

1. **State hypotheses:** H_0 and H_1
2. **Choose significance level:** α (e.g., 0.05)
3. **Select test statistic:** Z, t, or χ^2 depending on scenario
4. **Determine critical value(s)** or compute p-value
5. **Compute test statistic** from data
6. **Make decision:** Reject H_0 if test statistic in critical region (or $p < \alpha$)
7. **State conclusion** in engineering terms

15.11 Engineering Examples

Example 1: Is a Sensor Biased?

A sensor should read 0V with no input. 20 measurements give $\bar{X} = 0.012\text{V}$, $S = 0.03\text{V}$.

$H_0: \mu = 0$ (no bias), $H_1: \mu \neq 0$ (biased), $\alpha = 0.05$

$T = (0.012 - 0)/(0.03/\sqrt{20}) = 0.012/0.00671 = 1.789$

$t_{\{0.025, 19\}} = 2.093$. Since $|T| = 1.789 < 2.093 \rightarrow$ **Fail to reject H_0** .

p-value = $2 \cdot P(t(19) > 1.789) \approx 0.090$. Not significant at 5% level.

Conclusion: Insufficient evidence to conclude the sensor is biased.

Example 2: Did Manufacturing Process Change?

Before: $\mu_0 = 100\Omega$ (established). After modification: $n = 30$, $\bar{X} = 101.2\Omega$, $S = 3.5\Omega$.

$H_0: \mu = 100$, $H_1: \mu \neq 100$, $\alpha = 0.05$

$T = (101.2 - 100)/(3.5/\sqrt{30}) = 1.2/0.639 = 1.878$

$t_{\{0.025, 29\}} = 2.045$. $|T| = 1.878 < 2.045 \rightarrow$ **Fail to reject H_0** .

Example 3: Does New Algorithm Improve BER?

Old algorithm: $\text{BER}_0 = 10^{-3}$. New algorithm tested: $n = 50000$ bits, 38 errors.

$\text{BER}_{\text{new}} = 38/50000 = 7.6 \times 10^{-4}$

$H_0: p = 0.001$, $H_1: p < 0.001$ (improvement), $\alpha = 0.05$

$Z = (\hat{p} - p_0)/\sqrt{(p_0(1-p_0)/n)} = (0.00076 - 0.001)/\sqrt{(0.001 \times 0.999/50000)} = -0.00024/0.000141 = -1.70$

p-value = $P(Z < -1.70) = 0.0446 < 0.05 \rightarrow$ **Reject H_0** .

Conclusion: Evidence supports that the new algorithm has lower BER.

Example 4: Has Noise Power Increased?

Historical: $\sigma_0^2 = 0.04 \text{ V}^2$. New measurements: $n = 25$, $S^2 = 0.058 \text{ V}^2$.

$H_0: \sigma^2 = 0.04$, $H_1: \sigma^2 > 0.04$, $\alpha = 0.05$

$\chi^2 = (24)(0.058)/0.04 = 34.8$

$\chi^2_{\{0.05, 24\}} = 36.415$. Since $34.8 < 36.415 \rightarrow$ **Fail to reject H_0** .

15.12 MATLAB Examples**Example 1: One-Sample t-test**

```
%% Is the sensor biased? One-sample t-test
data = [0.012, -0.005, 0.031, 0.008, -0.012, 0.022, 0.015, ...
        0.003, 0.028, -0.001, 0.018, 0.009, 0.025, -0.008, ...
        0.014, 0.007, 0.020, 0.011, -0.003, 0.019];

mu_0 = 0; % Hypothesized mean (no bias)
[h, p_value, ci, stats] = ttest(data, mu_0);

fprintf('Test result: H = %d (1=reject H0)\n', h);
fprintf('p-value = %.4f\n', p_value);
fprintf('t-statistic = %.3f, df = %d\n', stats.tstat, stats.df);
fprintf('95% CI for  $\mu$ : [%.4f, %.4f]\n', ci);
```

MATLAB

Example 2: Two-Sample t-test

```
%% Compare two algorithms: is there a significant difference?
BER_algo1 = [0.0012, 0.0008, 0.0015, 0.0010, 0.0013, ...
             0.0011, 0.0009, 0.0014, 0.0012, 0.0010];
BER_algo2 = [0.0007, 0.0009, 0.0006, 0.0008, 0.0005, ...
             0.0008, 0.0007, 0.0006, 0.0009, 0.0007];

[h, p_value, ci, stats] = ttest2(BER_algo1, BER_algo2);
fprintf('Two-sample t-test:\n');
fprintf(' H = %d (1=reject H0: means are equal)\n', h);
fprintf(' p-value = %.4e\n', p_value);
fprintf(' t-stat = %.3f, df = %.1f\n', stats.tstat, stats.df);
fprintf(' Mean diff: %.4f\n', mean(BER_algo1) - mean(BER_algo2));
```

MATLAB

Example 3: Power Analysis

MATLAB

```
%% Power: How likely are we to detect a true difference?
mu_0 = 0;           % Null hypothesis mean
mu_true = 0.015;    % True mean (sensor has bias of 15mV)
sigma = 0.03;       % Known std dev
alpha = 0.05;

n_values = 5:5:100;
power = zeros(size(n_values));

for i = 1:length(n_values)
    n = n_values(i);
    % Critical value for two-sided test
    z_crit = norminv(1-alpha/2);
    % Power = P(reject | mu_true)
    % Reject when |Z| > z_crit, where Z ~ N((mu_true-mu_0)/(sigma/sqrt(n)), 1)
    ncp = (mu_true - mu_0) / (sigma/sqrt(n)); % Non-centrality parameter
    power(i) = 1 - normcdf(z_crit - ncp) + normcdf(-z_crit - ncp);
end

figure;
plot(n_values, power, 'b-o', 'LineWidth', 2);
xlabel('Sample Size n'); ylabel('Power (1 -  $\beta$ )');
title(sprintf('Power vs Sample Size (true bias = %.0f mV)', mu_true*1000));
grid on;
yline(0.8, 'r--', 'Desired Power = 0.8');
fprintf('Need n  $\approx$  %d for 80%% power\n', n_values(find(power >= 0.8, 1)));
```

15.13 Practice Problems**Problem 1**

A manufacturing spec requires mean resistance = 100Ω . A sample of 16 resistors gives $\bar{X} = 101.5\Omega$, $S = 3\Omega$. At $\alpha = 0.05$, is there evidence the process has drifted?

Solution $H_0: \mu=100$, $H_1: \mu \neq 100$. $T = (101.5-100)/(3/4) = 2.0$. $t_{0.025,15} = 2.131$. $|T|=2.0 < 2.131 \rightarrow$ Fail to reject. $p\text{-value} \approx 0.064 > 0.05$.

Problem 2

Average packet delay was 5ms. After network upgrade, 40 measurements give $\bar{X} = 4.6\text{ms}$, $S = 1.2\text{ms}$. Test if delay decreased at $\alpha = 0.01$.

Solution $H_0: \mu=5$, $H_1: \mu < 5$. $T = (4.6-5)/(1.2/\sqrt{40}) = -0.4/0.190 = -2.11$. $t_{0.01,39} \approx -2.426$. $T = -2.11 > -2.426 \rightarrow$ Fail to reject at 1% level. (Would reject at 5% since $t_{0.05,39} \approx -1.685$.)

Problem 3

Explain why “ $p = 0.04$ means there is only a 4% chance H_0 is true” is WRONG.

Solution The p-value is $P(\text{data this extreme} \mid H_0 \text{ true})$, NOT $P(H_0 \text{ true} \mid \text{data})$. It conditions on H_0 being true and asks about data extremity. To get $P(H_0 \text{ true} \mid \text{data})$ would require Bayesian analysis with a prior. The p-value is a property of the data-generating process under H_0 , not a posterior probability about H_0 .
Next Module: Module 16 — Engineering Statistical Applications